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Theoretical rate coefficients for allyl + HO_2 and allyloxy decomposition

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Abstract

The kinetics of the allyl + HO_2 bimolecular reaction, the thermal decomposition of C_3H_5OOH, and the unimolecular reactions of C_3H_5O are studied theoretically. High-level *ab initio* calculations of the C_3H_5OOH and C_3H_5O potential energy surfaces are coupled with RRKM master equation methods to compute the temperature- and pressure-dependence of the rate coefficients. Variable reaction coordinate transition state theory is used to characterize the barrierless transition states for the allyl + HO_2 and C_3H_5O + OH reactions. The predicted rate coefficients for allyl + HO_2 → C_3H_5OOH → products are in good agreement with experimental values. The calculations for allyl + HO_2 → C_3H_6 + O_2 underpredict the observed rate. The new rate coefficients suggest that the reaction of allyl + HO_2 will promote chain-branching significantly more than previous models suggest.
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Keywords: Allyl radical; HO_2 radical; Allyloxy radical; Barrierless kinetics; Master equation

1. Introduction

Bimolecular reactions involving HO_2 with other radicals are important in ignition chemistry, particularly at moderate temperatures and high pressures. Although not a chain-branching reaction, R + HO_2 reactions can promote ignition by converting less reactive radicals into more reactive radicals through the following sequence: R + HO_2 → RO + OH → aldehyde + H + OH. Alternatively, the hydroperoxyl radical can transfer the H-atom to the other radical, R + HO_2 → RH + O_2, and thus act as a chain-terminating reaction. The relative rates of these competing chain-propogating and chain-terminating reactions are critical for accurate modeling of ignition chemistry. A sensitivity analysis of the reaction rates in the counterflow ignition of methane confirm that CH_3 + HO_2 is among the most important reactions in methane ignition in the temperature range of 1100–1400 K [1]. Similarly, Heyberger et al. demonstrated that the induction period of propene ignition in the temperature range of 500–800 K is highly sensitive to allyl + HO_2 kinetics, and that omitting this reaction increased the induction period by an order of magnitude [2]. The allyl radical, C_3H_5, owing to its resonance stabilization, is less reactive than most radicals and therefore is present in high concentrations in many hydrocarbon flames, so its kinetics with HO_2 will be important for many larger hydrocarbons.

Allyl and HO_2 can react in one of four ways. First, three H-transfer reactions are possible. The allyl radical can abstract the H-atom from

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hydroperoxyl, forming propene and either singlet or triplet molecular oxygen. Alternatively, HO_2 can abstract the hydrogen from allyl, forming H_2O_2 + allene:

$$C_3H_5 + HO_2 \xrightarrow{k_{1a}} C_3H_6 + {}^3O_2 \quad (1a)$$

$$\xrightarrow{k_{1b}} C_3H_6 + {}^1O_2 \quad (1b)$$

$$\xrightarrow{k_{1c}} C_3H_4 + H_2O_2 \quad (1c)$$

HO_2 can add to either end of the allyl radical. The energized C_3H_5OOH adduct is either stabilized via collision or decomposes to two bimolecular channels:

$$C_3H_5 + HO_2 \xrightarrow{k_{2a}} C_3H_5OOH \quad (2a)$$

$$\xrightarrow{k_{2b}} CH_2CHCH_2O + OH \quad (2b)$$

$$\xrightarrow{k_{2c}} CH_2CHCHO + H_2O \quad (2c)$$

The O—O bond in the collisionally stabilized C_3H_5OOH is relatively weak, so this species can decompose to the same two channels:

$$C_3H_5OOH \xrightarrow{k_{3a}} CH_2CHCH_2O + OH \quad (3a)$$

$$\xrightarrow{k_{3b}} CH_2CHCHO + H_2O \quad (3b)$$

The sequence Reaction (2a) followed by (3b) is chain-terminating. Allyloxy, CH_2CHCH_2O, can undergo β-scission to form either vinyl + formaldehyde or H + acrolein. Alternatively it can undergo isomerization/decomposition reactions, three of which are listed below:

$$CH_2CHCH_2O \xrightarrow{k_{4a}} C_2H_3 + CH_2O \quad (4a)$$

$$\xrightarrow{k_{4b}} H + CH_2CHCHO \quad (4b)$$

$$\xrightarrow{k_{4c}} HCO + C_2H_4 \quad (4c)$$

$$\xrightarrow{k_{4d}} CH_2CH_2CHO \quad (4d)$$

$$\xrightarrow{k_{4e}} CH_2OCHCH_2 \quad (4e)$$

Despite the importance of the allyl + HO_2 reactions in hydrocarbon combustion, the literature on these reactions is sparse. The only experimental rate coefficients to our knowledge are from Lodhi & Walker [3,4]. In these experiments, 4,4-dimethyl-pent-1-ene thermally decomposed to t-butyl and allyl in the presence of O_2. Product yields were monitored by gas chromatography. A complex reaction mechanism is proposed in which the t-butyl radicals are assumed to react exclusively with O_2 to form iso-butene + HO_2. The mechanism included the following two allyl + HO_2 reactions:

$$C_3H_5 + HO_2 \xrightarrow{k_{1a}} C_3H_6 + {}^3O_2$$

$$C_3H_5 + HO_2 \xrightarrow{k_{eff}} CO + \text{products}$$

From measured yields of propene and CO they determined $k_{1a}/k_{eff} = 0.38 \pm 0.18$. Lodhi & Walker assumed that CO was formed in 100% yield from any species formed via $C_3H_5OOH^{\ddagger}$, i.e. $k_{eff} \approx k_2$. The individual rate constants for k_{1a} and k_{eff} were obtained from estimates of the initial allyl and HO_2 concentrations and other rate constants. These results were updated by Baulch et al. [5]. Both Lodhi & Walker and Baulch et al. assumed that the main source of CO is the production and subsequent oxidation of the vinyl radical. Tsang proposed a single step rate constant for allyl + HO_2 to OH + H + acrolein [6]. The mechanism of Heyberger et al. consisted of three reactions: H-abstraction, k_{1a}; formation of allylhydroperoxide, k_{2a}; and a single step for the decomposition of allylhydroperoxide to H + acrolein, $k_{3a–4b}$. The literature rates for Reactions 1a and 2 are summarized in Table 1. Rauk et al. [7] calculated the high-pressure rate coefficients for the decomposition of allyloxy to vinyl + formaldehyde and acrolein + H using CBS-RAD theory [8]. The related vinyl + formaldehyde → allyloxy potential energy surface was investigated by Xie et al. [9], using G3 [10] and CBS-QB3 [11] compound methods.

The kinetics of the allyl + HO_2 reaction and allyloxy decomposition are considerably more complex than the previous mechanisms suggest. Due to the barrierless entrance channel, the chemically-activated C_3H_5OOH adduct can react directly to either allyloxy + OH or acrolein + H_2O on the same time scale as the addition reaction. Thus, the rate coefficients for the reaction of allyl + HO_2 going to products and the subsequent decomposition of allyloxy will be a complex function of temperature and pressure, and the determination of these rate coefficients requires solving the master equation.

The methodology used in this paper is similar to that in Jasper et al. for the CH_3 + HO_2 reactions [12]. Variable reaction coordinate transition state theory (VRC-TST) is used to calculate the E, J-resolved microcanonical rate coefficients for the two barrierless reactions. Temperature and pressure dependence of the C_3H_5OOH and C_3H_5O addition/isomerization/decomposition rate coefficients are calculated by solving the master equation, using variational transition state theory and

Table 1
Rate coefficients for Reactions 1a and 2 at 753 K (10^{-11} cm^3 molecule^{-1} s^{-1}).

Source	k_{1a}	k_2
Lodhi and Walker [3] (indirect experiment)	0.31	0.74
Baulch [5] (review)	0.44	1.1
Tsang [6] (estimate)		1.6
Heyberger [2] (modeling)	0.31	0.85
Present work (theory)	0.01	1.2

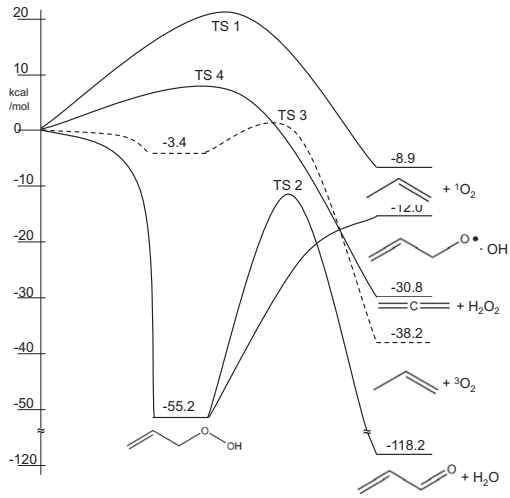


Fig. 1. Potential Energy Surface for allyl + HO_2.

rigid-rotor harmonic oscillator (RRHO) approximations.

2. Theory

2.1. Potential energy surfaces

For each species and transition state, the $\sim 3^n$ possible conformers – where n is the number of torsional modes – were calculated using the CBS-QB3 [11] method. The lowest energy conformer was selected, and this geometry was re-optimized using the B3LYP functional with the MG3S basis set, which was selected for the accuracy of its geometry and frequency calculations [13]. The DFT geometries were used for restricted QCISD(T)/cc-pVTZ and QCISD(T)/cc-pVQZ energies. The QCISD(T) complete basis set limit was extrapolated from the triple and quadruple zeta basis set calculations assuming an inverse power law [14,15]:

$$E_{\mathrm{CBS}} = E_{\mathrm{QZ}} + (E_{\mathrm{QZ}} - E_{\mathrm{TZ}})\frac{4^4}{5^4 - 4^4} \tag{1}$$

The QCISD(T)/CBS enthalpies at 0 K, including the B3LYP/MG3S zero point energies, for the $C_3H_6O_2$ potential energy surface (PES) are shown in Fig. 1 and listed in Tables 2 and 3; the

Table 2
Species for Allyl + HO_2 Potential Energy Surface.

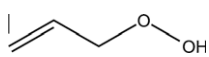
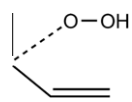
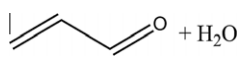
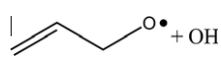
#	Species name	Structure	Relative energy (kcal/mol)	T1 diagnostic
R1	Allyl + HO_2		0.0	0.010, 0.037
W1	Allylhydroperoxide		−55.2	0.012
W2	vdW complex		−3.4	0.030
P1	Propene + $^3O_2$		−38.2	0.010, 0.007
P2	Propene + $^1O_2$		−8.9	0.010, 0.015
P3	Acrolein + H_2O		−118.2	0.015, 0.007
P4	Allyloxy + OH		−12.0	0.020, 0.007
P5	Allene + H_2O_2		−30.7	0.013, 0.011

Table 3
Transition States for Allyl + HO_2 Potential Energy Surface.

#	Reactions	Relative energy (kcal/mol)	ΔS_{298} (cal/mol/K)	T1 diagnostic
TS 1	R1 ↔ P2	22.8	−34.9	0.071
TS 2	W1 ↔ P3	−9.4	−40.9	0.022
TS 3	R1 ↔ P1	2.6	−43.5	0.035
TS 4	R1 ↔ P5	8.6	−9.2	0.041

corresponding values for the C_3H_5O PES are in Fig. 2 and Tables 4 and 5. Also included in Tables 2–5 are the QCISD(T)/cc-pVQZ T1 diagnostic [16] for the equilibrium and saddle point geometries. The T1 diagnostic is a measure of the importance of multireference effects. For closed shell species with a T1 diagnostic less than ~0.02, and for radicals with a T1 diagnostic less than ~0.03, the QCISD(T) energies are assumed to be accurate to ~1 kcal/mol, and multireference calculations are unnecessary [17]. CASPT2 [18] was used for the barrierless reactions. All DFT calculations were done using Gaussian03 [19]. All QCISD(T) and CASPT2 calculations were done using MOLPRO [20].

Allyl and HO_2 can react on both the singlet and triplet potential energy surfaces. On the triplet PES, indicated in Fig. 1 by the dashed lines, the two doublets form a weakly bound van der Waals complex. The allyl moiety abstracts the H-atom from the HO_2 via TS 3, leading to propene + $^3O_2$. On the singlet PES, HO_2 adds to either end of allyl, forming allylhydroperoxide, which dissociates to either allyloxy + OH or to acrolein + H_2O. Two possible H-transfer reactions can occur on the singlet surface: allyl + HO_2 → propene + $^1O_2$ via TS 1, and allyl + HO_2 → allene + H_2O_2 via TS 4. The T1 diagnostic for these reactions are 0.071 and 0.041, respectively, which implies that these transition states are ill-described by single-reference methods. Even if these barriers were off by 5 kcal/mol, these reactions would still be negligible compared to the other two reactions. Therefore, it is assumed that abstraction on the singlet surface can be safely neglected.

The salient features of the C_3H_5O PES are shown in Fig. 2. The lowest decomposition barrier for allyloxy is to acrolein + H via TS 7. Allyloxy can also undergo β-scission to form vinyl + formaldehyde via TS 5. Additionally, allyloxy can isomerize to the vinoxyl–methyl radical, via the oxiranyl-methyl radical, or it can isomerize to 2-formyl-ethyl, via an unusual 1,2-H shift, TS 6. All other competing barriers (e.g. 1,2-H shift to $CH_2CHCHOH$) were significantly higher. 2-formyl-ethyl dissociates to HCO + ethene via TS 9. Because this transition state is 13 kcal/mol lower in energy than the preceding isomerization barrier, it is possible for thermally activated allyloxy to decompose directly to HCO + ethene. Two other wells are available from 2-formyl-ethyl: 2-methyl-vinoxy and propionyl. These two wells and their decomposition channels (shown in the dotted lines in Fig. 2) were not included in the

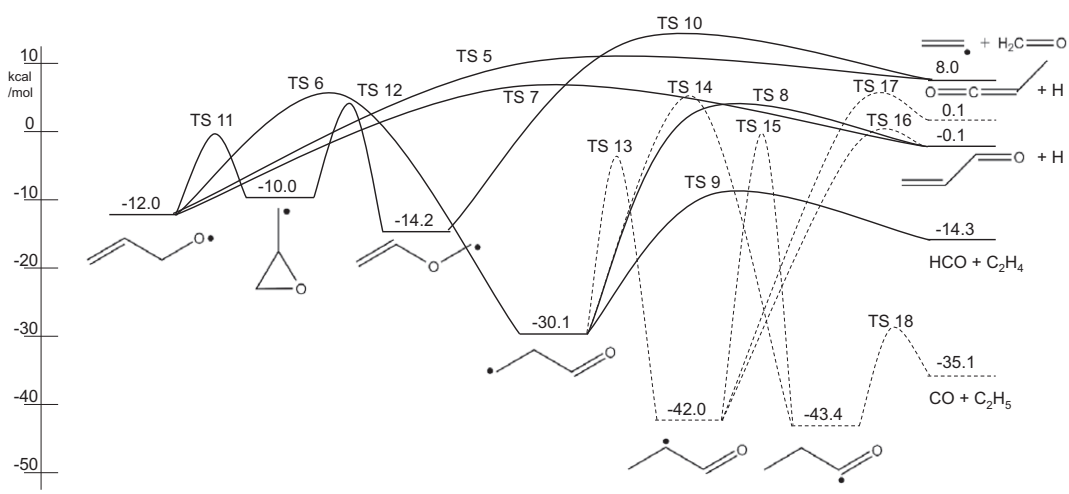
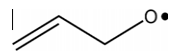
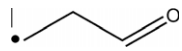
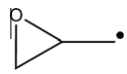
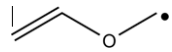
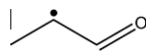
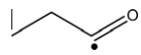
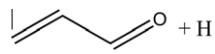
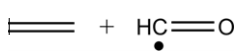
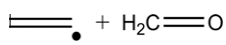
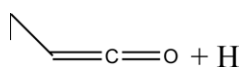
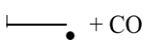


Fig. 2. C_3H_5O Potential Energy Surface (present work). The energies are relative to allyl + HO_2–OH. The dotted lines are omitted from the Master Equation calculations.

Table 4
Species for C_3H_5O Potential Energy Surface. Energies are relative to allyl + HO_2 – OH.

#	Species name	Structure	Relative energy (kcal/mol)	T1 diagnostic
W3	Allyloxy		−12.0	0.020
W4	2-Formyl–ethyl		−30.1	0.015
W5	Oxiranyl–methyl		−10.0	0.015
W6	Vinoxyl–methyl		−14.2	0.018
W7	2-Methyl-vinoxy		−42.0	0.020
W8	Propionyl		−43.4	0.019
P6	Acrolein + H		−0.1	0.015, 0.000
P7	C_2H_4 + HCO		−14.3	0.011, 0.025
P8	C_2H_3 + CH_2O		8.0	0.017, 0.016
P9	Methylketene + H		0.1	0.015, 0.000
P10	CO + C_2H_5		−35.1	0.018, 0.010

Master Equation calculations. The transition states leading to these isomers, TS 13 and TS 14, are more than 5 kcal/mol higher than the competing β-scission reaction, TS 9, and have a smaller A-factor. Sample calculations confirm these channels make a negligible contribution to the overall rate coefficients.

2.2. Variable reaction coordinate TST calculations

The two barrierless reactions, allyl + HO_2 ↔ allylhydroperoxide, and allylhydroperoxide ↔ allyloxy + OH, were calculated using variable reaction coordinate transition state theory [21–23]. These calculations were done using the computer code VaReCoF [24]. The degrees of freedom were separated into conserved and transitional modes. The conserved modes were the vibrational modes of the two fragments at their isolated geometries. The transitional modes, which are the coupled, anharmonic modes correlating with the relative orientation and separation of the two fragments, were calculated on-the-fly using multireference perturbation theory (CASPT2) [18]. The delocalized π-orbitals in the allyl radical require a minimum active space of three electrons in three orbitals (3e, 3o) [25]. Thus, for the allyl + HO_2 ↔ allylhydroperoxide reaction, a (4e, 4o) active space was chosen. For the allylhydroperoxide ↔ allyloxy + OH reaction the

Table 5
Transition States for C_3H_5O Potential Energy Surface. Energies are relative to allyl + HO_2–OH.

#	Reactions	Relative energy (kcal/mol)	ΔS_{298} (cal/mol/K)	T1 diagnostic
TS 5	W3 ↔ P8	10.3	−34.8	0.019
TS 6	W3 ↔ W4	6.1	−2.2	0.038
TS 7	W3 ↔ P6	6.7	0.6	0.019
TS 8	W4 ↔ P6	3.9	−1.1	0.017
TS 9	W4 ↔ P7	−6.7	0.6	0.023
TS 10	W6 ↔ P8	20.0	−34.1	0.025
TS 11	W3 ↔ W5	−4.5	−3.2	0.026
TS 12	W5 ↔ W6	3.3	−0.8	0.022
TS 13	W4 ↔ W7	−1.7	−3.7	0.018
TS 14	W4 ↔ W8	4.3	−3.9	0.026
TS 15	W7 ↔ W8	−0.4	−0.4	0.024
TS 16	W7 ↔ P6	1.4	2.1	0.016
TS 17	W7 ↔ P9	5.6	1.9	0.019
TS 18	W8 ↔ P10	−29.0	5.9	0.023

minimum active space was (6e, 4o), which accounts for the three electrons in two orbitals on each oxygen-centered radical [12]. For each reaction, two sets of CASPT2/cc-pVDZ geometry optimizations were performed along the minimum energy path (MEP) from 1.5 to 5 Å. In one set, all of the modes were relaxed; in the other set, the conserved modes were held fixed at their isolated geometries, and only the transitional modes were optimized. The CASPT2/CBS energy was extrapolated from subsequent CASPT2/aug-cc-pVDZ and CASPT2/aug-cc-pVTZ calculations using a formula similar to Eq. (1). The difference in energy between the constrained and relaxed MEP was added to the CBS correction and used as a 1-D correction to the interaction potential.

2.3. Master equation calculations

The time-dependent master equation was solved using the computer code VariFlex [26]. The torsional modes for each species and transition state were treated as hindered internal rotors. The barriers for internal rotation were calculated by a relaxed scan at the B3LYP/CBSB7 level. The resulting potential was fit to a Fourier series. A rigid-rotor harmonic oscillator model was assumed for all remaining degrees of freedom. In order to reduce the ambiguity regarding low-frequency vibrations and the corresponding torsional modes, the projection of the B3LYP/MG3S force-constant matrix was removed along the vectors corresponding to the torsional modes. This matrix was then diagonalized, which separated the rigid-rotor harmonic oscillator frequencies from the torsional modes [27]. These frequencies were then scaled by 0.9983, as recommended by Zhao et al. [13]. A Pitzer-Gwinn-like [28] approximation was made within Variflex for the hindered rotor partition function. The effective moment of inertia for each rotor was $I^{(2,3)}$, evaluated at the equilibrium geometry. Tunneling was included for all transition states by use of an Eckart approximation [29]. For energy transfer in the master equation, a single-exponential down model was used. Common literature values were used for the energy transfer parameter [30–32], with a $\langle\Delta E_{down}\rangle$ of 400 cm^{-1} $(T/298)^{0.8}$ for N_2. The collision frequency was estimated using a Lennard–Jones model, with LJ parameters of $\sigma = 4.42$ Å and $\varepsilon = 292$ cm^{-1} for C_3H_5OOH and $\sigma = 4.40$ Å and $\varepsilon = 280$ cm^{-1} for all C_3H_5O isomers. The LJ parameters for the wells were based on group additivity of the functional groups [33]. The N_2 bath gas LJ parameters, $\sigma = 3.74$ Å and $\varepsilon = 57.0$ cm^{-1}, were taken from literature values [34]. The $C_3H_6O_2$ and C_3H_5O master equation calculations were performed at 60 Torr of N_2 and temperatures between 400 and 2000 K.

3. Results and discussion

3.1. Allyl + HO_2 reaction

The calculated rate coefficients for the addition/dissociation pathway are shown in Fig. 3 and listed in Table 6. The high-pressure limit for the addition reaction, k_2^∞, is within a factor of 2 of the experimental data and is in excellent agreement with the updated results of Baulch. The error limits in all reported Arrhenius equations are the 95% confidence intervals and do not include uncertainties in the thermochemistry. At 400 K and 60 Torr of N_2, roughly 20% of the C_3H_5OOH adduct is collisionally stabilized to allylhydroperoxide; for temperatures greater than 600 K more than 99% of the energized C_3H_5OOH adduct goes directly to allyloxy + OH. In contrast, if the pressure is increased to 10 atm of N_2 (not shown), allylhydroperoxide is the dominant product channel until 800 K. The acrolein + H_2O channel is negligible at all temperatures and pressures. Roaming radical reactions [35], which could

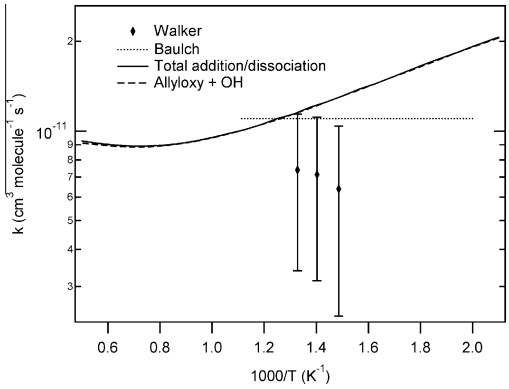


Fig. 3. Allyl + HO_2 → $CH_2CHCH_2OOH^{\ddagger}$ → products rate coefficients in 60 Torr of N2.

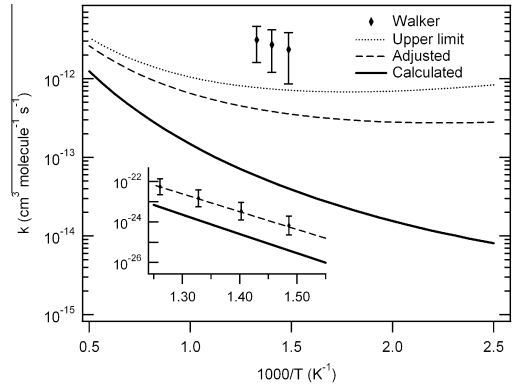


Fig. 4. Allyl + HO_2 → propene + $^3O_2$ rate coefficients in 60 Torr of N_2. The inset is for the reverse reaction: propene + $^3O_2$ → Allyl + HO_2.

conceivably increase the acrolein + H_2O channel yield, were not considered in the present work.

The rate for the abstraction reaction on the triplet surface is shown in Fig. 4 and listed in Table 6. The computed rate coefficient is nearly a factor of 30 slower than that inferred by Lodhi & Walker from experimental data. The T1 diagnostic for H-abstraction on the triplet surface is 0.035, which suggests minor uncertainty due to multireference effects. The transition state geometry was re-optimized using the (6e, 6o)-CASPT2/cc-PVTZ method, and the CASPT2/CBS energy was extrapolated from subsequent CASPT2 calculations

Table 6
Computed rate coefficients in 60 Torr of N_2. The units are: A in cm^3, molecule^{-1}, s^{-1}; T in Kelvin; E_a in Kelvin. The superscript $^\infty$ refers to the high-pressure limit. The rate expression is: $k = A\,(T/298)^n \exp[-E_a/T]$.

Reaction			A	n	E_a
Allyl + HO_2	$\overset{k_{1a}}{\rightarrow}$	Propene + $^3O_2$	$(9.3 \pm 0.1) \times 10^{-15}$	2.7	490 ± 10
	$\overset{k_{1a}^{adjusted}}{\rightarrow}$	Propene + $^3O_2$	$(7.4 \pm 0.1) \times 10^{-15}$	2.8	-1100 ± 10
	$\overset{k_{1a}^{upperlimit}}{\rightarrow}$	Propene + $^3O_2$	$(7.6 \pm 0.1) \times 10^{-15}$	2.8	-1540 ± 10
	$\overset{k_{2a}^{\infty}}{\rightarrow}$	Allylhydroperoxide	$(1.7 \pm 0.1) \times 10^{-12}$	0.6	-1060 ± 10
	$\overset{k_{2a}}{\rightarrow}$	Allylhydroperoxide	$(8.7 \pm 0.9) \times 10^{-14}$	−3.7	-650 ± 50
	$\overset{k_{2b}}{\rightarrow}$	Allyloxy + OH	$(2.0 \pm 0.5) \times 10^{-12}$	0.5	-990 ± 10
	$\overset{k_{2c}}{\rightarrow}$	Acrolein + H_2O	$(9.5 \pm 0.2) \times 10^{-18}$	3.4	-1390 ± 10
Allylhydroperoxide	$\overset{k_{3a}}{\rightarrow}$	Allyloxy + OH	$(5.9 \pm 0.3) \times 10^{16}$	−7.2	19560 ± 110
	$\overset{k_{3a}^{\infty}}{\rightarrow}$	Allyloxy + OH	$(3.1 \pm 0.1) \times 10^{16}$	−1.5	21580 ± 10
	$\overset{k_{3b}}{\rightarrow}$	Acrolein + H_2O	$(1.4 \pm 0.1) \times 10^{12}$	−7.0	20740 ± 50
Allyloxy	$\overset{k_{4a}}{\rightarrow}$	Vinyl + CH_2O	$(2.1 \pm 0.1) \times 10^{10}$	−1.0	9380 ± 50
	$\overset{k_{4a}^{\infty}}{\rightarrow}$	Vinyl + CH_2O	$(2.7 \pm 0.1) \times 10^{14}$	0.0	12030 ± 10
	$\overset{k_{4b}}{\rightarrow}$	Acrolein + H	$(1.8 \pm 0.1) \times 10^{11}$	−1.7	7880 ± 50
	$\overset{k_{4b}^{\infty}}{\rightarrow}$	Acrolein + H	$(3.5 \pm 0.1) \times 10^{12}$	1.0	9340 ± 20
	$\overset{k_{4c}}{\rightarrow}$	Ethene + HCO	$(3.9 \pm 0.1) \times 10^{10}$	−1.8	6860 ± 40
	$\overset{k_{4d}}{\rightarrow}$	2-Formyl-ethyl	$(6.2 \pm 0.1) \times 10^{9}$	−3.5	6740 ± 10
	$\overset{k_{4e}}{\rightarrow}$	Vinoxyl–methyl	$(8.5 \pm 0.2) \times 10^{11}$	−3.5	6950 ± 40

using the cc-pVTZ and cc-pVQZ basis sets. The CASPT2/CBS barrier is −1.0 kcal/mol lower than the QCI/CBS barrier. Thus, the uncertainty in this transition state is assumed to be ±3 kcal/mol. Stothard & Walker measured the rate coefficient for the reverse reaction, k_{-1a}, $C_3H_6 + {}^3O_2 \rightarrow$ allyl + HO_2 [36]. The present thermochemistry was used to calculate the rate coefficient for this reaction, shown in the inset in Fig. 4. The calculated rate coefficient using the QCI/CBS energy is a factor of five too slow. If the barrier is decreased by 3.0 kcal/mol (i.e. $\Delta E_0 = 37.8$ kcal/mol relative to propene + ${}^3O_2$, instead of $\Delta E_0 = 40.8$ kcal/mol), the calculated rate coefficient agrees with the data to within 25% (see inset). If this new barrier is used for the forward reaction (i.e. $\Delta E_0 = -0.4$ instead of 2.6 kcal/mol), the calculated rate coefficient remains too slow by a factor of 10 (the dashed line in Fig. 4). If the uncertainty in the thermochemistry for allyl + HO_2 relative to propene + ${}^3O_2$ is ±1 kcal/mol, and if ΔH_0 is thereby decreased from −38.2 to −39.2 kcal/mol, while holding the transition state barrier at $\Delta E_0 = 37.8$ kcal/mol relative to propene + ${}^3O_2$, the adjusted rate coefficient is within a factor of three of the experimental data. This adjusted rate coefficient (the dotted line in Fig. 4) is an upper bound for allyl + $HO_2 \rightarrow C_3H_6 + {}^3O_2$. To match the data to within 25%, ΔH_0 would need to be decreased by 3.3 kcal/mol, which correlates to an unrealistically large decrease of 6 kcal/mol in the forward barrier height. Since the experimental rate coefficients for k_{-1a} are less error-prone than the forward rate, and since the calculated thermochemistry for the stable species should be accurate to ±1 kcal/mol, the present results suggest that the previous rate coefficient for allyl + $HO_2 \rightarrow C_3H_6 + {}^3O_2$ is at least a factor of three too fast. In all likelihood, it is an order of magnitude off, and may possibly be off by as much as a factor of 30. The experimental results suggest that a larger mechanism, possibly one including heterogeneous effects, is needed to explain some of the observed species.

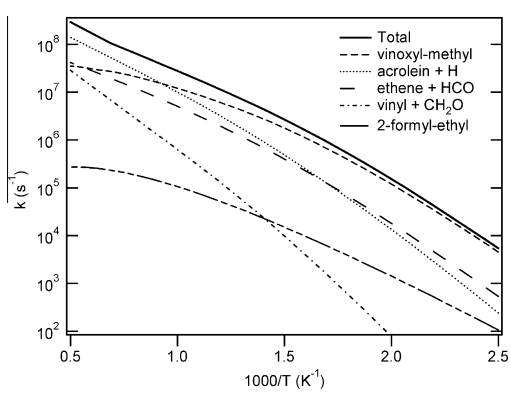


Fig. 5. Computed rate coefficients for allyloxy decomposition in 60 Torr of N_2.

3.2. Allylhydroperoxide dissociation

At high pressures and low-to-moderate temperatures, the activated C_3H_5OOH adduct is collisionally stabilized to allylhydroperoxide. The stabilized product will then dissociate to form allyloxy + OH. The rate coefficients for allylhydroperoxide dissociation are listed in Table 6. At 60 Torr of N_2, the rate constant is roughly an order of magnitude slower than the high pressure limit. The other two unimolecular channels available to allylhydroperoxide are four orders of magnitude slower than dissociation to allyloxy + OH at all temperatures and pressures.

3.3. Allyloxy kinetics

Whether formed directly via chemical activation or indirectly via collisional stabilization and subsequent thermal dissociation of allylhydroperoxide, the dominant product in the allyl + HO_2 reaction is allyloxy + OH. The results of master equation simulations of allyloxy dissociation in 60 Torr of N_2 are shown in Fig. 5 and listed in Table 6. At this pressure, the two dissociation channels are several orders of magnitude below their high-pressure limits. The largest rate coefficient is isomerization to vinoxyl–methyl. Kinetic simulations suggest that allyloxy and vinoxyl–methyl rapidly equilibrate, followed by decomposition. Below 600 K, HCO + ethene is computed to be the dominant bimolecular product. Above 600 K, acrolein + H is the dominant bimolecular product, but the HCO + ethene channel is within a factor of three for all temperatures. This important channel has been neglected in all previous mechanisms, even though ethene production was observed experimentally. In the temperature range of the experiments, the rate of production of vinyl + formaldehyde is less than 2% of the total rate of allyloxy decomposition to bimolecular products. Thus, it is unlikely that the dominant source of CO in the experiments is from this channel. Instead, the CO probably resulted from (i) HCO decomposition and (ii) H-abstraction from acrolein to form an unsaturated acyl radical, $CH_2{=}CHC({=}O)$, which decomposes to vinyl + CO [37–39]. Although allyl + HO_2 can go directly to allyloxy + OH via chemical activation, it is unlikely that the allyloxy fragment will have sufficient internal energy to fragment immediately on the same time scale, since the barrier for decomposition to acrolein + H is ~7 kcal/mol above allyl + HO_2 in energy. Thus, modeling the kinetics by a single step reaction that omits the allyloxy intermediate is not advised.

Transition state TS 6 determines the branching fraction between HCO + ethene versus acro-

lein + H. The T1 diagnostic for this transition state is 0.038. Consequently, there is some uncertainty in this barrier height due to multireference effects. Subsequent (5e, 5o)-CASPT2/CBS//CASPT2/aug-cc-pVDZ calculations lowered the barrier from $\Delta E_0 = 18.1$ kcal/mol to $\Delta E_0 = 16.6$ kcal/mol, and CI + QC/CBS corrections to the CASPT2 geometry increased the barrier from $\Delta E_0 = 18.1$ kcal/mol to $\Delta E_0 = 20.5$ kcal/mol. In order to test the sensitivity of the total branching fractions to this transition state, the RRKM/ME results were repeated with the barrier height altered ±2 kcal/mol. Increasing the barrier by 2 kcal/mol increased the rate of acrolein + H by ~20% and decreased the rate of HCO + ethene by a factor of 5. Similarly, decreasing the barrier by 2 kcal/mol decreased the rate of acrolein + H by a factor of 2 and increased the rate of HCO + ethene by a factor of 3.

4. Conclusions

A theoretical investigation of the kinetics of the allyl + HO_2 bimolecular reaction and subsequent kinetics have been performed over a wide range of temperatures and pressures relevant for combustion modeling. This study presents for the first time a set of comprehensive pressure-dependent rate coefficients. Rate coefficients were calculated using *ab initio* theory, variable reaction coordinate transition state theory, variational transition state theory, and master equation simulations. The present calculations are in good agreement with available experimental data for the addition reaction. The calculations for the abstraction reaction on the triplet surface are slower than the experimentally determined rates by a factor of 30, but agree with the reverse reaction within a factor of five. Some of this discrepancy may be attributed to minor multireference effects in the transition state, but mostly it is due to the indirect methods used to measure the rate constant. At lower pressures and/or higher temperatures, most of the allyl + HO_2 goes directly to allyloxy + OH via chemical activation. Rate coefficients for allyloxy decomposition are also presented. Allyloxy will decompose to H + acrolein, HCO + ethene, and vinyl + CH_2O. The prompt formation of OH, and the slower rate of formation of propene + O_2, suggest that this reaction can promote chain-branching significantly more than previous models suggest.

Acknowledgements

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Appendix A. Supplementary data

The B3LYP/MG3S geometries for the transition states in Tables 3 and 5 are provided. Pressure-dependent rate coefficients were calculated between 400 and 2000 K and 0.1–100 atm of N_2, and are included in a CHEMKIN-compatible mechanism. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.054.

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